

CO1-AT2

CSA1701

NAME : SHAIK THOUHID

REG NO: 192421280

ARTIFICIAL INTELLIGENCE & SEARCH ALGORITHMS

Comprehensive Assignment Solutions | Backtracking, Grid Search & Autonomous Rescue Agents

Question 1: Hospital Shift Scheduling via Backtracking Search

Problem Formulation:

- Variables: Doctors D1, D2, D3
- Domains: Shifts {Morning (M), Afternoon (A), Night (N)} mapped chronologically as $M < A < N$
- Constraints:
 1. $D1 \neq \text{Night}$
 2. $\text{Shift}(D2) < \text{Shift}(D3)$
 3. All shifts distinct (One doctor per shift, bijection between $\{D1, D2, D3\}$ and $\{M, A, N\}$)
 4. $D3 \neq \text{Morning}$
 5. Every doctor assigned exactly one shift

Step-by-Step Backtracking Search Execution

1. Step 1 — Assign Shift to D1:

- Try D1 = Morning (M):
 - Check Constraints: $D1 \neq \text{Night}$ (Satisfied).
 - Remaining available shifts for D2, D3: {Afternoon (A), Night (N)}.

2. Step 2 — Assign Shift to D2 (given D1 = M):

- Try D2 = Afternoon (A):
 - Check Constraints: Distinct shift from D1 (Satisfied).
 - Remaining available shift for D3: {Night (N)}.

3. Step 3 — Assign Shift to D3 (given D1 = M, D2 = A):

- Try D3 = Night (N):

- Check Constraint iv ($D3 \neq M$): Satisfied ($N \neq M$).
- Check Constraint ii ($D2 < D3$): $A < N$ is TRUE (Afternoon precedes Night).
- Check Constraint iii (All distinct): $\{M, A, N\}$ are all distinct.
- All constraints satisfied! Complete valid assignment found.

Backtracking Tree Evaluation & Verification Table

Branch / Path	Assignments	Constraint Checks	Status
Path 1 (Target)	D1=M, D2=A, D3=N	$D1 \neq N$ (OK), $D3 \neq M$ (OK), $D2 < D3$ ($A < N$: OK), All Distinct	VALID (Solution)
Path 2 (Alt Branch)	D1=M, D2=N, D3=A	$D2 < D3$ fails (Night does not precede Afternoon)	BACKTRACK
Path 3 (Alt Branch)	D1=A, D2=M, D3=N	$D1 \neq N$ (OK), $D3 \neq M$ (OK), $D2 < D3$ ($M < N$: OK), All Distinct	VALID (Solution)
Path 4 (Pruned)	D1=N, ...	$D1 \neq \text{Night}$ violated immediately at depth 1	PRUNED

Final Valid Hospital Shift Schedule

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FINAL SHIFT ASSIGNMENT:
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  • Morning Shift   (M) : Doctor D1
  • Afternoon Shift (A) : Doctor D2
  • Night Shift     (N) : Doctor D3
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(Note: Alternative valid assignment: D1 = Afternoon, D2 = Morning, D3 = Night)
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Question 2: 5×5 Grid Navigation via Best-First / A* Search

Environment Setup & Specifications:

- Grid Size: 5×5 coordinates (0..4, 0..4)
- Start (S): (0, 0) | Goal (G): (4, 4)

- Obstacles (X): (1, 1), (1, 2), (2, 3), (3, 1)
- Heuristic Function: Manhattan Distance $h(x, y) = |x - 4| + |y - 4|$
- Step Cost: $g(\text{node}) = 1$ per orthogonal step
- Algorithm Evaluation Function: $f(n) = g(n) + h(n)$ [A* Search / Informed Best-First]

Visual Representation of 5×5 Grid

	Col 0	Col 1	Col 2	Col 3	Col 4
0	[S]	[]	[]	[]	[]
1	[]	[X]	[X]	[]	[]
2	[]	[]	[]	[X]	[]
3	[]	[X]	[]	[]	[]
4	[]	[]	[]	[]	[G]

Step-by-Step Priority Queue Trace

Step	Expanded Node	Successors Generated (x, y)	$f(n) = g(n) + h(n)$	Priority Queue / Open List (Sorted by f)
0	Init	(0,0)	$g=0, h=8, f=8$	[(0,0): $f=8$]
1	(0,0)	(1,0), (0,1)	(1,0): $g=1, h=7, f=8$ (0,1): $g=1, h=7, f=8$	[(1,0): $f=8, (0,1): f=8$]
2	(1,0)	(2,0) [Note: (1,1) is Obstacle]	(2,0): $g=2, h=6, f=8$	[(0,1): $f=8, (2,0): f=8$]
3	(2,0)	(3,0), (2,1)	(3,0): $g=3, h=5, f=8$ (2,1): $g=3, h=5, f=8$	[(0,1): $f=8, (3,0): f=8, (2,1): f=8$]
4	(2,1)	(2,2) [Note: (3,1),(1,1) Obstacles]	(2,2): $g=4, h=4, f=8$	[(2,2): $f=8, (0,1): f=8, (3,0): f=8$]
5	...	(2,2) -> (3,2) -> (4,2) -> (4,3) -> (4,4)	Progresses directly toward G with minimal $f(n)$	[(4,4): $f=8$]

Optimal Path & Total Movement Cost

Optimal Sequence of Grid Coordinates:

(0, 0) → (1, 0) → (2, 0) → (2, 1) → (2, 2) → (3, 2) → (4, 2) → (4, 3) → (4, 4)

Cost Calculation:

- Total Moves = 8 steps
- Cost per Move = 1
- Total Optimal Path Cost = 8 units

Question 3: Autonomous Disaster Rescue Robot Agent

Part a) Analysis of Constraints on Decision-Making

- 1. 4-Directional Movement:** Restricts action space to {Up, Down, Left, Right}, dictating grid-based successor generation.
- 2. Obstacle Blocks:** Introduces hard environmental constraints; blocked cells cannot be expanded, requiring dynamic path re-routing.
- 3. Partial Observability (Adjacent Sensing Only):** The agent operates with an incomplete map. Unvisited cells are unknown, preventing full offline path computation upfront and necessitating local state discovery.
- 4. Variable Path Costs (Regular = 1, Risky = 3):** Differentiates safe traverse (cost=1) from high-risk structural zones (cost=1+2=3). Forces the agent to balance path distance against traversal risk.
- 5. Dynamically Updated Knowledge (Online Search):** The agent must interleave planning and execution, updating its internal map representation upon each perception step.

Part b) Proposed Solution Approach: LRTA* / Online Uniform Cost Search

Selected Search Framework: Learning Real-Time A* (LRTA*) with Dynamic Cost Mapping (Online UCS).

Algorithmic Execution Steps:

- 1. Perception Phase:** Sense adjacent cells (Up, Down, Left, Right) from current position s . Identify walls, open paths, and risky zones.
- 2. Knowledge Base Update:** Update local grid map with observed cell traversal costs: regular = 1, risky = 3, obstacle = ∞ .
- 3. Heuristic / Cost Evaluation:** For each adjacent non-blocked neighbour s' , calculate estimated total cost $f(s') = g(s') + h(s')$.
- 4. Action Selection:** Move to the neighbouring cell s' that minimizes the estimated remaining cost.
- 5. Value Learning & Loop:** Update the heuristic cost estimate of current state s based on neighbour values: $H(s) \leftarrow \max(H(s), \min_{s'} [c(s, s') + H(s')])$. Repeat until Goal G is reached.

Part c) Demonstration of Core AI Concepts

AI Concept	Theoretical Framework	Application in Rescue Robot
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Scenario		
Intelligent Agent	PEAS Paradigm (Performance, Environment, Actuators, Sensors)	P: Maximize survivor safety, minimize cost. E: Grid building with collapse zones. A: Wheeled/tracked motors (4- way). S: Adjacent distance/risk sensors.
Rationality	Expected Utility Maximization	The robot makes optimal local choices based on current percept history, choosing low-risk paths over shorter high-risk paths when total expected cost is lower.
Environment Type	Task Environment Classification	• Partially Observable (senses 1- hop) • Dynamic / Stochastic (structural risks) • Sequential (actions affect future states) • Discrete (grid rooms)

Part d) Justification of the Selected Search Strategy

• **Infeasibility of Standard Offline Search (A* / Dijkstra):** Standard A* requires complete global knowledge of the graph prior to execution. In a partially observable disaster environment, the robot cannot construct an offline plan because unmapped obstacles and unknown risky zones would invalidate pre-computed paths.

• **Advantage of Online Search (LRTA* / Dynamic UCS):** Online search interleaves computation with real-time sensing and movement. It guarantees completeness in finite state spaces and avoids infinite loops by updating state memory (learning), ensuring safe, optimal, and adaptive navigation toward survivor G.